

Informe de experimento: llm_only_forced_nr

Generado el 2026-09-16 12:53 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio llm_only_forced_nr (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_validation_nr/gpt-4o-mini/llm_only_forced_nr
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-4o-mini
- Métodos: llm_only_forced:nr
- Casos: case14
- Requests: generadas, N=40 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=10, parameterized=10, multistep=10, ambiguous=10
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 12:47
- Costo total: 0.0449 USD (40 items; costo medio 0.00112 USD/item)
- Items totales: 40; ok=36; con error=4; tokens totales=147092
- Solved global: 5.0 % (2/40)

Errores por tipo:

Tipo	Items	Métodos afectados
json_parse_failed	4	llm_only_forced:nr (4)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método llm_only_forced:nr

System prompt —860 caracteres; strategy=nr, forced=True, sin herramientas

You are a power system analysis expert. You need to calculate AC power flow results based on test system data provided by the user.

Important Constraints:

- You must strictly output JSON (no extra text, no Markdown blocks).
- Do not output NaN/Infinity; all values must be finite real numbers.
- You must compute and report your best numerical estimate for every bus voltage and every branch flow. `converged` must be true and every array must be complete (one entry per bus and per branch). Do not leave arrays empty and do not state that you cannot compute.

Reasoning mode: you MAY write a short step-by-step reasoning in plain text before the answer. The final JSON object must be the LAST thing in the response, inside a single ```json fenced block, and it must be the only JSON object in the response. Do not use curly braces in the reasoning text.

User prompt (primer item) —6381 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

System base: 100 MVA. Every per-unit quantity below (r_pu, x_pu, b_pu) is defined on this base. Line and transformer impedances (r_pu, x_pu, b_pu) are already in per unit on this base, not ohms per km; do not convert them further. Branch thermal ratings are not defined for this case, so branch loading is undefined: report loading_percent as 0 for every branch.

You are a power systems expert. Based on the following IEEE case14 test system data, calculate the AC power flow results.

System Data (from pandapower network tables)

Node Data (Bus Table: net.bus)

	name	type	zone	in_service
0	1	b	1.0	True
1	2	b	1.0	True

```

2    3    b  1.0      True
3    4    b  1.0      True
4    5    b  1.0      True
5    6    b  1.0      True
6    7    b  1.0      True
7    8    b  1.0      True
8    9    b  1.0      True
9   10    b  1.0      True
10  11    b  1.0      True
... (3 líneas omitidas de 15)

```

Load Data (Load Table: net.load)

	bus	p_mw	q_mvar	in_service
0	2	21.473798	12.567615	True
1	3	85.726702	17.290948	True
2	4	45.568126	-3.717902	True
3	5	6.887322	1.449962	True
4	6	11.740497	7.861940	True
5	9	32.229269	18.135792	True
6	10	9.714580	6.260507	True
7	11	3.424253	1.761044	True
8	12	6.539657	1.715320	True
9	13	14.757610	6.340306	True
10	14	15.783969	5.296634	True

Generator Data (Generator Table: net.gen)

	bus	p_mw	vm_pu	min_p_mw	max_p_mw	min_q_mvar	max_q_mvar	in_service
0	2	41.63323	1.045	0.0	140.0	-40.0	50.0	True
1	3	0.00000	1.010	0.0	100.0	0.0	40.0	True
2	6	0.00000	1.070	0.0	100.0	-6.0	24.0	True
3	8	0.00000	1.090	0.0	100.0	-6.0	24.0	True

Balanced Node/External Grid (External Grid Table: net.ext_grid)

	bus	vm_pu	va_degree	in_service
0	1	1.06	0.0	True

Line Data (Line Table: net.line)

	from_bus	to_bus	r_pu	x_pu	b_pu	parallel	in_service
0	1	2	0.01938	0.05917	0.0528	1	True
1	1	5	0.05403	0.22304	0.0492	1	True
2	2	3	0.04699	0.19797	0.0438	1	True
3	2	4	0.05811	0.17632	0.0340	1	True
4	2	5	0.05695	0.17388	0.0346	1	True
5	3	4	0.06701	0.17103	0.0128	1	True
6	4	5	0.01335	0.04211	0.0000	1	True
7	6	11	0.09498	0.19890	0.0000	1	True
8	6	12	0.12291	0.25581	0.0000	1	True
9	6	13	0.06615	0.13027	0.0000	1	True
10	9	10	0.03181	0.08450	0.0000	1	True

... (4 líneas omitidas de 16)

Transformer Data (Transformer Table: net.trafo)

	hv_bus	lv_bus	r_pu	x_pu	shift_degree	tap_side	tap_neutral	tap_min	tap_max	tap_step_percent	in_service
0	4	7	0.0	0.20912	0.0	hv	0.0	NaN	NaN	2.2	True
1	4	9	0.0	0.55618	0.0	hv	0.0	NaN	NaN	3.1	True
2	5	6	0.0	0.25202	0.0	hv	0.0	NaN	NaN	6.8	True
3	7	8	0.0	0.17615	0.0	None	NaN	NaN	NaN	NaN	True
4	7	9	0.0	0.11001	0.0	None	NaN	NaN	NaN	NaN	True

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Reasoning Instructions

Solve this with the Newton-Raphson power-flow procedure, worked by hand in the reasoning text before the final JSON:

- Build the bus admittance matrix Ybus (per unit, on the stated system base) from the r_pu/x_pu/b_pu of every line and transformer (series admittance $1/(r_{pu}+jx_{pu})$, plus the shunt charging b_pu split half to each end bus).
 - Classify each bus as slack (ext_grid), PV (gen) or PQ (load only); PV/slack voltage magnitudes are the given setpoints, not unknowns.
 - Flat-start the unknowns: angle 0 at every non-slack bus, magnitude 1.0 p.u. at every PQ bus.
 - At each iteration: compute the power mismatch (Delta P at every non-slack bus, Delta Q at every PQ bus) from the current voltage estimate, form the Jacobian, solve for the angle/magnitude correction, and update the voltage estimate. Show the mismatch vector and the updated voltages at each iteration explicitly.
 - Repeat until the largest mismatch is below $1e-3$ p.u. (report the iteration count), or state explicitly that you are not iterating further and why.
- Do not skip straight to a final answer without showing at least one iteration's mismatch and update; a plausible-looking answer with no shown iteration is not an acceptable substitute.

Output Requirements

Please calculate and return the following results (JSON format):

- Voltage magnitude (p.u.) and phase angle (degrees) for each node
- Active power (MW) and load factor (%) at the beginning of each branch

- For lines: `line\_id` uses the row index of `net.line`
 - For transformers: `line\_id` uses the row index of 100000 + `net.trafo`

3) Total generation (MW), total load (MW), total losses (MW)
 4) Are there any voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)?

Please strictly follow the following JSON schema output (all fields are complete and of correct type):

```
{
  "converged": true/false,

  "bus_voltages": [{"bus_id": 1, "vm_pu": 1.0, "va_deg": 0.0}, ...],

  "line_flows": [{"line_id": 0, "p_from_mw": 0.0, "loading_percent": 0.0}, ...],

  "total_generation_mw": 0.0,

  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

[Important] bus_id must use the `name` column of net.bus (the usual IEEE/MATPOWER 1..N numbering), not the row index of net.bus.

3. Resultados agregados

Columnas del protocolo (medias por método $\times$ caso, del `scoreboard_per_case`), en los 4 grupos de `metricas_pfagent_definicion`
utilidad de tarea Form. (formulación exacta de las tool calls), V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

Por método $\times$ caso

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Abst.	V pass	\$
llm_only_forced:nr	case14	40	-	2.82e-02	37.85	5.0 (2/40)	100.0 (36/36)	1.22e+018.4	-	-	n/a	3677	-	0.0 (0/40)	0.0 (0/40)	0.0449

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
llm_only_forced:nr	n/a	n/a	n/a	n/a

Solved por dificultad (% , items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
llm_only_forced:nr	0.0 (0/10)	0.0 (0/10)	20.0 (2/10)	0.0 (0/10)

Tipos de error de formulación (conteo de items)

Method	unparsed
llm_only_forced:nr	4

Errores de ejecución por método (conteo de items con error)

Method	json_parse_failed
llm_only_forced:nr	4

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de `traces/` cuando existe.

Método `llm_only_forced:nr`

Ejemplo: resuelto (solved=True)

```
request (case14-multistep-002-s0, case14, difficulty=multistep, seed=1627694679): Load the 14-bus test case (case14), disconnect the line
↳ between bus 6 and bus 13, disconnect the line between bus 12 and bus 13, and finally rerun the power flow, then list every line whose
↳ loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,
↳ "to_bus": 13}) -> run_powerflow({})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
To solve the power flow problem for the given IEEE 14-bus test system, we will follow the outlined steps:

1. Build the Bus Admittance Matrix (Ybus)**: We will calculate the admittance for each line and transformer, taking into account the
↳ series impedance and shunt admittance. The admittance for a line is given by  $Y = 1/(r + jx)$  and the shunt admittance is  $b/2$  at each bus.

2. Classify Buses**: Identify the slack bus (external grid), PV buses (generators), and PQ buses (loads).

3. Flat Start**: Initialize the voltage magnitudes and angles for the non-slack buses.

4. Iterate**: Calculate power mismatches, form the Jacobian matrix, and update voltage estimates until convergence is achieved.

5. Report Results**: After convergence, we will report the bus voltages, branch flows, t ...[2314 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=2.65e-02; F_MAE=38.70; faithful=1.6 %; solved=True (answer_ok); flags=-; error=-; tokens=3597;
↳ cost_usd=0.0011; wall=15.4 s
```

Ejemplo: no resuelto

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
To solve the power flow problem for the given IEEE case14 test system, we will follow the Newton-Raphson method step-by-step.

1. Build the Bus Admittance Matrix (Ybus)**:
- For each line and transformer, we calculate the series admittance  $Y_{series} = \frac{1}{r + jx}$  and the shunt admittance  $Y_{shunt} = \frac{b}{2}$ .
↳ We will sum the admittances for each bus to form the Ybus matrix.

2. Classify Buses**:
- Bus 1 is the slack bus (external grid).
- Buses 2, 3, 6, and 8 are PV buses (generators).
- The remaining buses (4, 5, 7, 9, 10, 11, 12, 13, 14) are PQ buses (loads).

3. Flat Start**:
- Initialize voltage magnitudes at 1.0 p.u. for all PQ buses and angles at 0 degrees for all non-slack buses.

4. Iterate**:
- Calculate power mismatches (Delta P ...[3148 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=2.67e-02; F_MAE=35.45; faithful=0.0 %; solved=False (voltage_error); flags=-; error=-; tokens=4059;
↳ cost_usd=0.0014; wall=21.6 s
```

Ejemplo: primer item

```
request (case14-ambiguous-007-s0, case14, difficulty=ambiguous, seed=2073320062): Load the 14-bus test case (case14) and disconnect the
↳ line between bus 6 and bus 13 (1-based).
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
To solve the power flow problem for the given IEEE 14-bus test system, we will follow the steps outlined in the reasoning instructions.

1. Build the Bus Admittance Matrix (Ybus)**: We will calculate the admittance for each line and transformer, taking into account the
↳ series impedance and shunt admittance. The Ybus matrix will be constructed based on the connections between the buses.

2. Classify Each Bus**: We will identify the slack bus (bus 1), PV buses (buses 2, 3, 6, and 8), and PQ buses (the remaining buses with
↳ loads).

3. Flat Start**: We will initialize the voltage magnitudes and angles. The slack bus will have a voltage magnitude of 1.06 p.u. and an
↳ angle of 0 degrees. The PV buses will have their specified voltage magnitudes, and the PQ buses will start with a voltage ma ...[3252
↳ chars más]
```

```

--- Scoring ---
formulation=n/a (exact=None); V_MAE=2.00e-02; F_MAE=40.57; faithful=19.2 %; solved=False (voltage_error); flags=-; error=-; tokens=3955;
↪ cost_usd=0.0013; wall=21.7 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- LLM-only: V_MAE entre 2.82e-02 (llm_only_forced:nr/case14) y 2.82e-02 (llm_only_forced:nr/case14) p.u. sobre los items que devolvieron números.
- 0.0 % (0/40) de abstención en LLM-only (respuesta con `converged=false` o arrays vacíos).
- En LLM-only, 0 items no resueltos por `convergence_mismatch` y 34 por `voltage_error`.
- 4 items con `json_parse_failed`: la respuesta LLM-only no fue JSON parseable.
- Solved de `llm_only_forced:nr`: 5.0 % (2/40).
- Ningún item reclamó éxito sobre un fallo (`claimed_success_on_failure=0`).
- Trazabilidad de números (Faith.) media en LLM-only: 8.4 % (sin tools, todo número es no trazable salvo los copiados de la request).
- Costo total 0.0449 USD; el método más caro fue `llm_only_forced:nr` con 0.0449 USD (100 %).

6. Archivos

Rutas absolutas bajo `llm_only_forced_nr.report.rescored.json` (cuando existe) es el archivo leído para las métricas.

```

report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2]
↪ /power-flow-agent/benchmarks/results_validation_nr/gpt-4o-mini/llm_only_forced_nr/report.json    [leído:
↪ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGr]
↪ ids-Code-v2/power-flow-agent/benchmarks/results_validation_nr/gpt-4o-mini/llm_only_forced_nr/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGri]
↪ ds-Code-v2/power-flow-agent/benchmarks/results_validation_nr/gpt-4o-mini/llm_only_forced_nr/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrid]
↪ s-Code-v2/power-flow-agent/benchmarks/results_validation_nr/gpt-4o-mini/llm_only_forced_nr/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-Fo]
↪ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_nr/gpt-4o-mini/llm_only_forced_nr/scoreboard_per_case.json
scoreboard.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-Fo]
↪ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_nr/gpt-4o-mini/llm_only_forced_nr/scoreboard.rescored.json
scoreboard_per_case.rescored.json:
↪ /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/p]
↪ ower-flow-agent/benchmarks/results_validation_nr/gpt-4o-mini/llm_only_forced_nr/scoreboard_per_case.rescored.json
traces (40 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-Sm]
↪ artGrids-Code-v2/power-flow-agent/benchmarks/results_validation_nr/gpt-4o-mini/llm_only_forced_nr/traces

```